

Multivariate GBM and Dirichlet Portfolio Weights

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Model correlated assets.** Write and solve a multivariate GBM model whose matrix square root generates a specified log-return covariance rate.
- **Estimate covariance.** Construct, annualize, and validate a sample covariance matrix while distinguishing covariance from correlation.
- **Sample portfolio weights.** Use a Dirichlet distribution to explore the long-only allocation simplex without confusing random sampling with optimization.

1 Why Separate Asset Models Are Insufficient

Single-asset GBM models can be simulated independently, but an independently simulated portfolio erases co-movement. Firms respond together to market news, sectors share inputs, and hedges move in opposing directions. Portfolio uncertainty depends not only on each asset's volatility but also on every pairwise covariance.

Let $m \geq 2$ be the number of assets. Collect the prices in $\mathbf{S}_t \in \mathbb{R}_{>0}^m$, arithmetic GBM drifts in $\boldsymbol{\mu} \in \mathbb{R}^m$, and independent Wiener processes in $\mathbf{W}_t \in \mathbb{R}^m$. A multivariate GBM (MGBM) uses a loading matrix $A \in \mathbb{R}^{m \times m}$:

$$\frac{dS_{i,t}}{S_{i,t}} = \mu_i dt + \sum_{j=1}^m A_{ij} dW_{j,t}, \quad i \in \{1, \dots, m\}. \quad (1)$$

The diffusion covariance rate is $C := AA^\top$, whose entries have units years⁻¹. Its diagonal entries satisfy $C_{ii} = \sigma_i^2$, while C_{ik} records instantaneous log-return covariance per unit time. Applying Itô's lemma to $\log S_{i,t}$ gives the expected continuously compounded growth rate

$$\bar{g}_i := \mathbb{E}[g_{i,t,\Delta t}] = \mu_i - \frac{1}{2}C_{ii}, \quad \mu_i = \bar{g}_i + \frac{1}{2}C_{ii}. \quad (2)$$

Thus μ_i is reserved for the arithmetic drift in the price SDE, while $g_{i,t,\Delta t} := (\Delta t)^{-1} \log(S_{i,t+\Delta t}/S_{i,t})$ is the realized growth rate. Both have units years⁻¹, but they are different quantities.

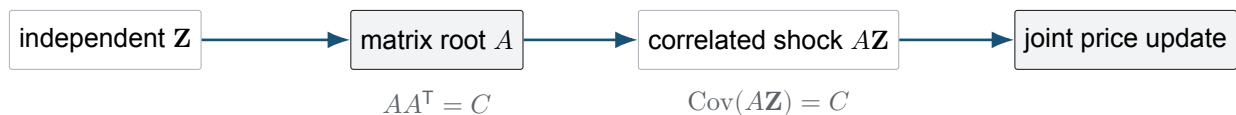


Figure 1: A matrix square root converts independent standard-normal shocks into shocks with covariance C . The same shock vector must drive all assets on one simulated path to preserve their joint dependence.

2 Exact Transition and Covariance

The logarithm of each component receives its own Itô correction, while the off-diagonal covariance enters through the common shock vector (Prop. 2.1).

Proposition 2.1. Exact constant-parameter MGBM transition

Let $A \in \mathbb{R}^{m \times m}$, $C := AA^\top$, $\boldsymbol{\mu} \in \mathbb{R}^m$, and $\Delta t > 0$ years. Define $\bar{g}_i := \mu_i - C_{ii}/2$. For independent standard normal shocks $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_m)$, the exact one-step solution of Eq. 1 is

$$S_{i,t+\Delta t} = S_{i,t} \exp \left[\bar{g}_i \Delta t + \sqrt{\Delta t} (A\mathbf{Z})_i \right]. \quad (3)$$

Define the dimensionless log-return vector $\mathbf{r}_t := \log(\mathbf{S}_{t+\Delta t} \oslash \mathbf{S}_t)$ and the realized growth-rate vector $\mathbf{g}_t := \mathbf{r}_t / \Delta t$. Then

$$\mathbf{g}_t = \bar{\mathbf{g}} + \frac{1}{\sqrt{\Delta t}} A\mathbf{Z}, \quad \mathbb{E}[\mathbf{g}_t] = \bar{\mathbf{g}} = \boldsymbol{\mu} - \frac{1}{2} \text{diag}(C), \quad \text{Cov}(\mathbf{r}_t) = C\Delta t, \quad \text{Cov}(\mathbf{g}_t) = \frac{C}{\Delta t}. \quad (4)$$

Equation Eq. 4 gives the scaling rule. Writing $\Sigma_r := \text{Cov}(\mathbf{r}_t)$ and $\Sigma_g := \text{Cov}(\mathbf{g}_t)$,

$$\Sigma_r = (\Delta t)^2 \Sigma_g, \quad C = \frac{\Sigma_r}{\Delta t} = \Delta t \Sigma_g. \quad (5)$$

The three matrices are not interchangeable: Σ_g is appropriate for a growth-rate portfolio objective, while C drives GBM shocks. Multiplying a daily log-return covariance by “252” estimates C ; it does not estimate Σ_g .

3 Estimating and Validating the Covariance Matrix

Let $G \in \mathbb{R}^{n \times m}$ store n observed growth-rate vectors by rows, and let $\bar{\mathbf{g}} \in \mathbb{R}^m$ be the column mean. With $\mathbf{1}_n$ the length- n vector of ones, define the centered matrix $\tilde{G} := G - \mathbf{1}_n \bar{\mathbf{g}}^\top$. The unbiased growth-rate covariance is

$$\hat{\Sigma}_g := \frac{1}{n-1} \tilde{G}^\top \tilde{G}. \quad (6)$$

It is symmetric and positive semidefinite: for any $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{x}^\top \hat{\Sigma}_g \mathbf{x} = \|\tilde{G}\mathbf{x}\|_2^2 / (n-1) \geq 0$. For GBM simulation, use $\hat{C} = \Delta t \hat{\Sigma}_g$.

Covariance depends on units and volatility scale. Correlation normalizes those effects:

$$\rho_{ik} := \frac{\Sigma_{ik}}{\sqrt{\Sigma_{ii} \Sigma_{kk}}}, \quad -1 \leq \rho_{ik} \leq 1 \quad (7)$$

when both variances are positive. Zero correlation means zero linear covariance under the chosen sample and horizon; it does not generally imply statistical independence.

A Cholesky factor is convenient when $\hat{\Sigma}$ is positive definite. Duplicated assets, short samples, or exact linear dependence can make it only semidefinite, causing an unpivoted Cholesky routine to fail. Eigenvalue or singular-value square roots can handle semidefinite matrices. Arbitrarily adding diagonal “jitter” changes the model and must be documented rather than hidden.

Sampling error is often the dominant problem. Pairwise estimates can shift across windows, and a matrix with $m(m+1)/2$ distinct entries becomes difficult to estimate when m is large relative to n . Shrinkage and factor models trade some flexibility for greater stability.

COMPANION NOTEBOOK**L5a: Computing and Analyzing a Covariance Matrix** →

Center a multi-asset growth-rate matrix, calculate $\widehat{\Sigma}_g$, convert it to the GBM covariance rate $\widehat{C}$, and verify numerical invariants.

4 Buy-and-Hold Portfolio Wealth

Let total initial wealth be $W_0 > 0$ dollars and let $\mathbf{w} \in \mathbb{R}^m$ contain initial dollar fractions. For a fully invested long-only portfolio, $w_i \geq 0$ and $\mathbf{1}^\top \mathbf{w} = 1$. Fractional-share holdings are $n_i = w_i W_0 / S_{i,0}$. If no rebalancing occurs, portfolio wealth is

$$W_t = \sum_{i=1}^m n_i S_{i,t} = W_0 \sum_{i=1}^m w_i \frac{S_{i,t}}{S_{i,0}}. \quad (8)$$

The initial weights generally drift as relative prices change. A constant-weight portfolio is a different, dynamically rebalanced strategy with additional turnover and cost assumptions.

COMPANION NOTEBOOK**L5b: Correlated MGBM Portfolio Simulation** →

Generate correlated asset paths, construct a buy-and-hold portfolio, and compare simulated wealth with observed and benchmark wealth.

5 Dirichlet Sampling Explores the Simplex

The feasible long-only weight set is the $(m-1)$ -dimensional simplex $\Delta^{m-1} := \{\mathbf{w} \in \mathbb{R}^m : w_i \geq 0, \sum_i w_i = 1\}$. The Dirichlet distribution generates random points on its interior and is a useful design-of-experiments tool (Defn. 5.1).

Definition 5.1. Dirichlet portfolio weights

Let $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_m)$ with every concentration parameter $\alpha_i > 0$, and let $\alpha_0 := \sum_{i=1}^m \alpha_i$. A random vector $\mathbf{W} \sim \text{Dirichlet}(\boldsymbol{\alpha})$ lies on Δ^{m-1} and has mean and component variance

$$\mathbb{E}[W_i] = \frac{\alpha_i}{\alpha_0}, \quad \text{Var}(W_i) = \frac{\alpha_i(\alpha_0 - \alpha_i)}{\alpha_0^2(\alpha_0 + 1)}. \quad (9)$$

For equal $\alpha_i = 1$, the distribution is uniform with respect to volume on the simplex; it is not uniform in each component separately.

Small equal concentrations place more mass near sparse corners, while large equal concentrations cluster weights near equal allocation. Unequal concentrations encode a directional sampling preference. None of these draws is “optimal” unless an objective and constraints are subsequently evaluated. Dirichlet sampling explores candidate portfolios; L5b solves a minimum-variance optimization problem.

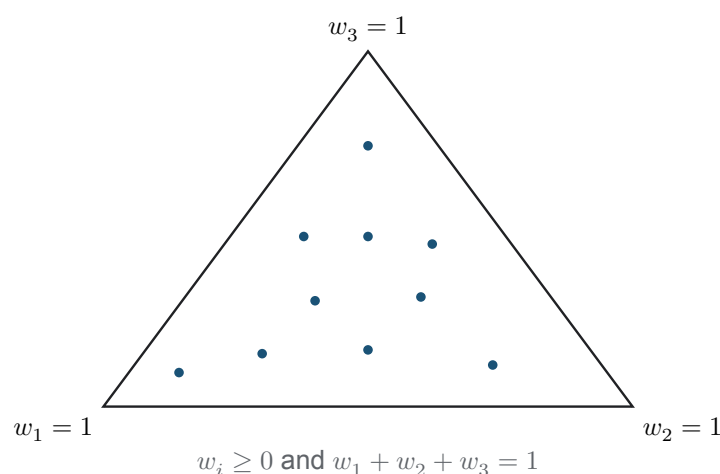


Figure 2: The three-asset long-only simplex. Dirichlet draws provide valid candidate allocations throughout the triangle; their density depends on the concentration vector α .

KEY TAKEAWAYS

In this lecture, we coupled GBM assets through a covariance matrix, built correlated shocks, and used Dirichlet draws to explore valid portfolio weights.

- **Drift and growth are distinct.** The price SDE uses μ , while realized growth is $g_t = (\Delta t)^{-1} \log(S_{t+\Delta t} \oslash S_t)$ and $\mathbb{E}[g_t] = \mu - \frac{1}{2} \text{diag}(C)$.
- **Covariance drives joint risk.** A matrix square root maps independent shocks into correlated log returns with covariance $\Sigma \Delta t$.
- **Estimates require structural checks.** A covariance estimate must be symmetric and positive semidefinite, and its scaling, window, and regularization must be documented.

Sampling is not optimization

Dirichlet draws explore the long-only simplex; concentration parameters control exploration but do not encode a minimum-risk objective. Next, turn estimated means and covariance into the global minimum-variance portfolio, efficient frontier, capital allocation line, and tangency portfolio.

ADDITIONAL READING

- Harry Markowitz, “Portfolio Selection,” *Journal of Finance* 7(1), 77–91 (1952).
- Robb J. Muirhead, *Aspects of Multivariate Statistical Theory* (1982), for covariance matrices and multivariate normal transformations.
- Thomas M. Cover and Joy A. Thomas, *Elements of Information Theory*, 2nd ed. (2006), Appendix, for the Dirichlet family and simplex-valued distributions.

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